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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 25th July Morning Shift

JEE Main Question Paper with Answer Key

25 Jul 2021	90	19	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	23 questions
Mathematics	22 questions
Physics	26 questions
Out of syllabus	19 questions
Answer key	at the end

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Chemistry 23 questions

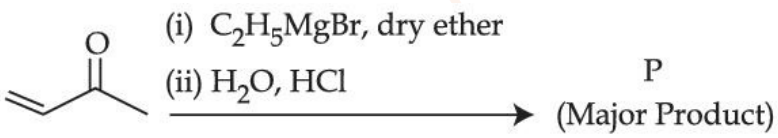
Q1. MCQ (Single Correct Answer)

Which one of the following species responds to an external magnetic field?

(A) $[Fe(H_2O)_6]^{3+}$ (B) $[Ni(CN)_4]^{2-}$ (C) $[Co(CN)_6]^{3-}$ (D) $[Ni(CO)_4]$

Open solution

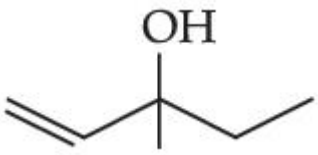
Q2. MCQ (Single Correct Answer)



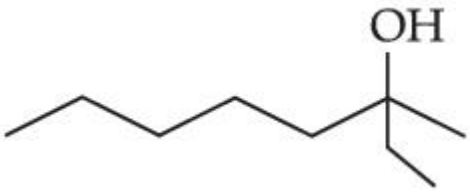
Open solution

Consider the above reaction, the major product 'P' is :

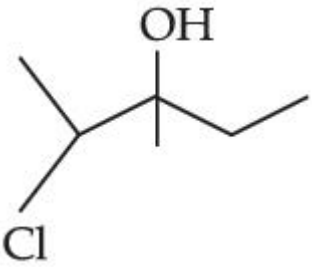
(A)



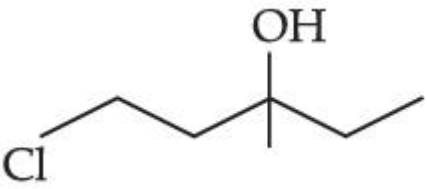
(B)



(C)



(D)



Q3. MCQ (Single Correct Answer)

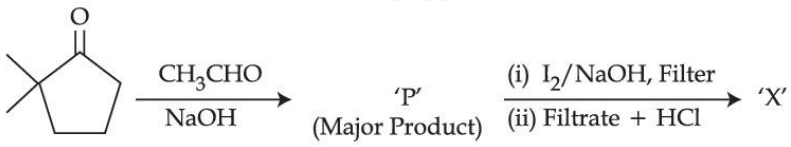
The water soluble protein is :

(A) Fibrin (B) Albumin (C) Myosin (D) Collagen

Open solution

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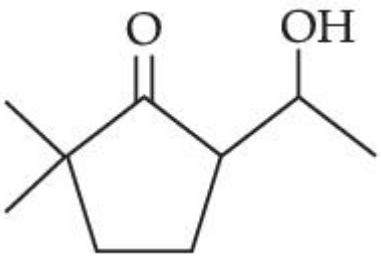
Q4. MCQ (Single Correct Answer)



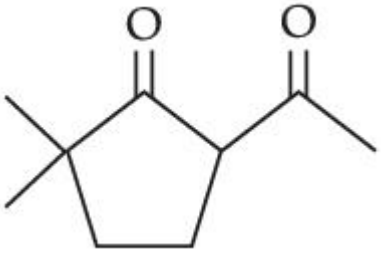
Open solution

Consider the given reaction, the product 'X' is :

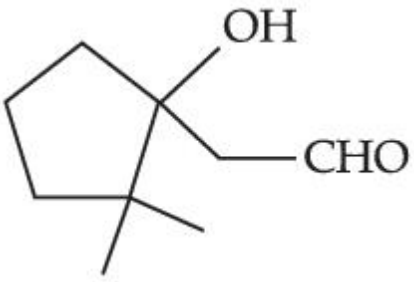
(A)



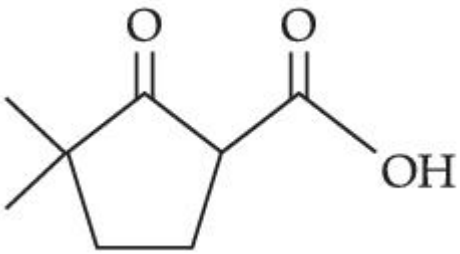
(B)



(C)



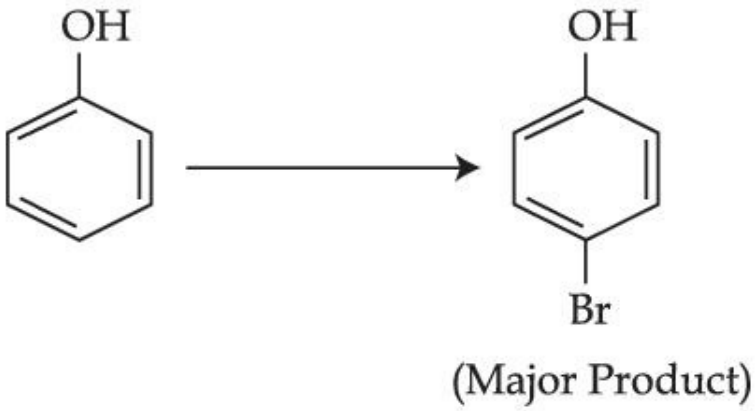
(D)



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Q5. MCQ (Single Correct Answer)

Open solution



(Major Product)

The given reaction can occur in the presence of :

(a) Bromine water

(b) Br_2 in CS_2, 273 K

(c) $Br_2/FeBr_3$

(d) Br_2 in $CHCl_3$, 273 K

Choose the correct answer from the options given below :

(A) (b) and (d) only

(B) (a) and (c) only

(C) (b), (c) and (d) only

(D) (a), (b) and (d) only

Q6. MCQ (Single Correct Answer)

Open solution

Given below are two statements, one is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A) : Gabriel phthalimide synthesis cannot be used to prepare aromatic primary amines.

Reason (R) : Aryl halides do not undergo nucleophilic substitution reaction.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) true but (R) is not the correct explanation of (A)

(B) (A) is false but (R) is true.

(C) Both (A) and (R) true and (R) is correct explanation of (A).

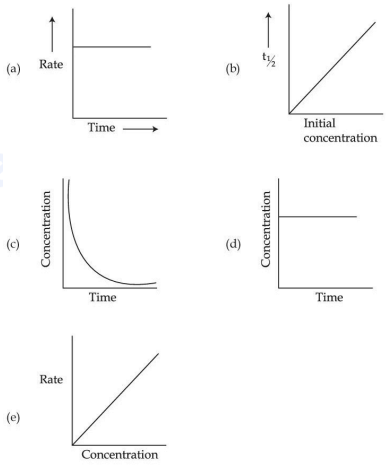
(D) (A) is true but (R) is false.

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Q7. MCQ (Single Correct Answer)

Open solution

For the following graphs,



Choose from the options given below, the correct one regarding order of reaction is :

(A) (b) Zero order (c) and (e) First order

(B) (a) and (b) Zero order (e) First order

(C) (b) and (d) Zero order (e) First order

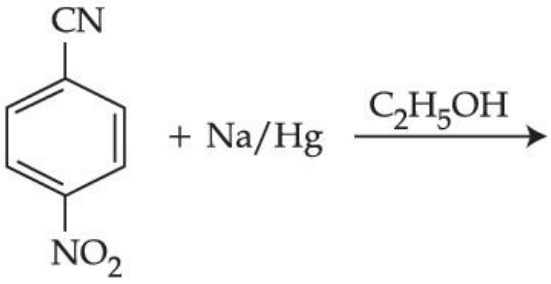
(D) (a) and (b) Zero order (c) and (e) First order

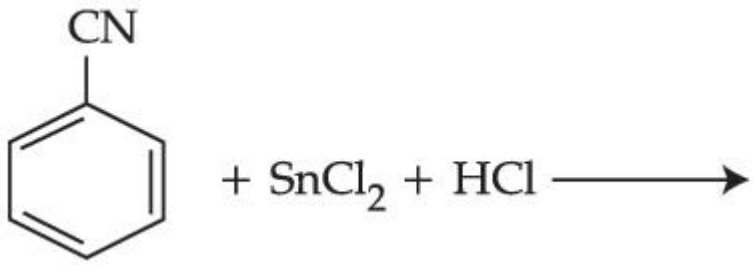
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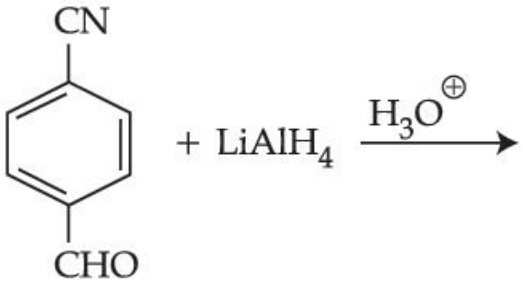
Q8. MCQ (Single Correct Answer)

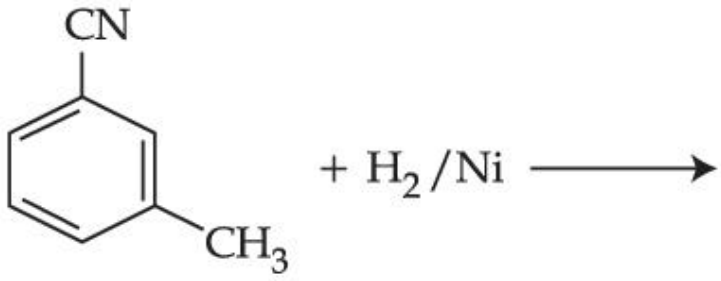
Open solution

Which one of the products of the following reactions does not react with Hinsberg reagent to form sulphonamide?

(A) 

(B) 

(C) 

(D) 

Q9. MCQ (Single Correct Answer)

Open solution

The ionic radii of K^+, Na^+, Al^{3+} and Mg^{2+} are in the order:

(A) $Na^+ < K^+ < Mg^{2+} < Al^{3+}$

(B) $Al^{3+} < Mg^{2+} < K^+ < Na^+$

(C) $Al^{3+} < Mg^{2+} < Na^+ < K^+$

(D) $K^+ < Al^{3+} < Mg^{2+} < Na^+$

Q10. MCQ (Single Correct Answer)

Open solution

Which one of the following compounds of Group-14 elements is not known?

(A) $[GeCl_6]^{2-}$

(B) $[Sn(OH)_6]^{2-}$

(C) $[SiCl_6]^{2-}$

(D) $[SiF_6]^{2-}$

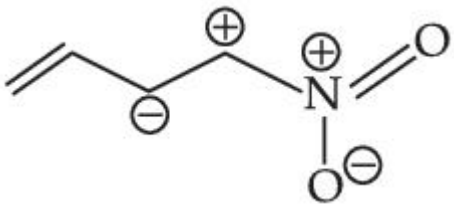
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Q11. MCQ (Single Correct Answer)

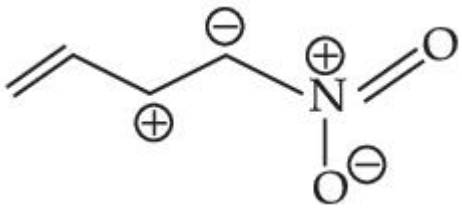
Open solution

Which one among the following resonating structures is not correct?

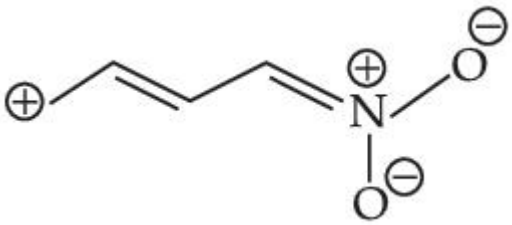
(A)



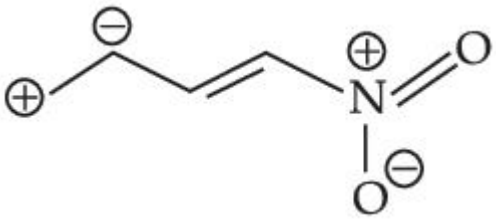
(B)



(C)



(D)



Q12. MCQ (Single Correct Answer)

Open solution

The correct order of following 3d metal oxides, according to their oxidation numbers is :

(a) CrO_3, (b) Fe_2O_3, (c) MnO_2, (d) V_2O_5, (e) Cu_2O

(A) (d) > (a) > (b) > (c) > (e)

(B) (a) > (c) > (d) > (b) > (e)

(C) (a) > (d) > (c) > (b) > (e)

(D) (c) > (a) > (d) > (e) > (b)

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Q13. MCQ (Single Correct Answer)

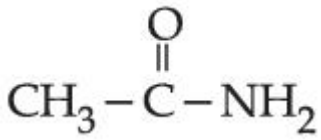
Which one of the following compounds will liberate CO_2, when treated with $NaHCO_3$?

Open solution

(A) $(CH_3)_3\overset{\oplus}{N}H\overset{\ominus}{Cl}$

(B) $(CH_3)_4\overset{\oplus}{N}\overset{\ominus}{O}H$

(C)



(D) CH_3NH_2

Q14. MCQ (Single Correct Answer)

Open solution

An organic compound 'A' C_4H_8 on treatment with $KMnO_4/H^+$ yields compound 'B' C_3H_6O.

Compound 'A' also yields compound 'B' on ozonolysis. Compound 'A' is :

(A) 2-Methylpropene

(B) 1-Methylcyclopropane

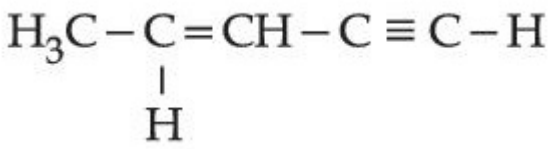
(C) But-2-ene

(D) Cyclobutane

Q15. Numerical

Open solution

The number of sigma bonds in



is _____.

Q16. Numerical

Open solution

Three moles of AgCl get precipitated when one mole of an octahedral co-ordination compound with empirical formula $CrCl_3.3NH_3.3H_2O$ reacts with excess of silver nitrate. The number of chloride ions satisfying the secondary valency of the metal ion is _____.

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Q17. Numerical

A source of monochromatic radiation of wavelength 400 nm provides 1000 J of energy in 10 seconds. When this radiation falls on the surface of sodium, x $\times$ 10^{20} electrons are ejected per second. Assume that wavelength 400 nm is sufficient for ejection of electron from the surface of sodium metal. The value of x is _____. (Nearest integer)

(h = 6.626 $\times$ 10^{-34} Js)

Open solution

Q18. Numerical

CO_2 gas is bubbled through water during a soft drink manufacturing process at 298 K. If CO_2 exerts a partial pressure of 0.835 bar then x m mol of CO_2 would dissolve in 0.9 L of water. The value of x is _____. (Nearest integer)

(Henry's law constant for CO_2 at 298 K is 1.67 $\times$ 10^3 bar)

Open solution

Q19. Numerical

For the reaction

$A + B \rightleftharpoons 2C$

the value of equilibrium constant is 100 at 298 K. If the initial concentration of all the three species is 1 M each, then the equilibrium concentration of C is x $\times$ 10^{-1} M. The value of x is _____. (Nearest integer)

Open solution

Q20. Numerical

Consider the cell at 25°C

$Zn \mid Zn^{2+}(aq), (1M) \parallel Fe^{3+}(aq), Fe^{2+}(aq) \mid Pt(s)$

The fraction of total iron present as Fe^{3+} ion at the cell potential of 1.500 V is x $\times$ 10^{-2}. The value of x is _____. (Nearest integer)

(Given : $E^0_{Fe^{3+}/Fe^{2+}} = 0.77V$, $E^0_{Zn^{2+}/Zn} = -0.76V$)

Open solution

Q21. Numerical

At 298 K, the enthalpy of fusion of a solid (X) is 2.8 kJ mol^{-1} and the enthalpy of vaporisation of the liquid (X) is 98.2 kJ mol^{-1}. The enthalpy of sublimation of the substance (X) in kJ mol^{-1} is _____. (in nearest integer)

Open solution

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Q22. Numerical

When 10 mL of an aqueous solution of Fe^{2+} ions was titrated in the presence of dil H_2SO_4 using diphenylamine indicator, 15 mL of 0.02 M solution of $K_2Cr_2O_7$ was required to get the end point. The molarity of the solution containing Fe^{2+} ions is $x \times 10^{-2}$ M. The value of x is _____. (Nearest integer)

Open solution

Q23. Numerical

Consider the complete combustion of butane, the amount of butane utilized to produce 72.0 g of water is _____ $\times 10^{-1}$ g. (in nearest integer)

Open solution

Mathematics 22 questions

Q24. MCQ (Single Correct Answer)

Let $f(x) = 3\sin^4 x + 10\sin^3 x + 6\sin^2 x - 3$, $x \in \left[-\frac{\pi}{6}, \frac{\pi}{2}\right]$. Then, f is :

(A) increasing in $\left(-\frac{\pi}{6}, \frac{\pi}{2}\right)$

(B) decreasing in $\left(0, \frac{\pi}{2}\right)$

(C) increasing in $\left(-\frac{\pi}{6}, 0\right)$

(D) decreasing in $\left(-\frac{\pi}{6}, 0\right)$

Open solution

Q25. MCQ (Single Correct Answer)

Let S_n be the sum of the first n terms of an arithmetic progression. If $S_{3n} = 3S_{2n}$, then the value of $\frac{S_{4n}}{S_{2n}}$ is :

(A) 6 (B) 4 (C) 2 (D) 8

Open solution

Q26. MCQ (Single Correct Answer)

The locus of the centroid of the triangle formed by any point P on the hyperbola $16x^2 - 9y^2 + 32x + 36y - 164 = 0$, and its foci is :

(A) $16x^2 - 9y^2 + 32x + 36y - 36 = 0$

(B) $9x^2 - 16y^2 + 36x + 32y - 144 = 0$

(C) $16x^2 - 9y^2 + 32x + 36y - 144 = 0$

(D) $9x^2 - 16y^2 + 36x + 32y - 36 = 0$

Open solution

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Q27. MCQ (Single Correct Answer)

Open solution

Let f : R $\rightarrow$ R be defined as

$$f(x)=\begin{cases}\frac{\lambda|x^2-5x+6|}{\mu(5x-x^2-6)}, & x<2\\ e^{\frac{\tan(x-2)}{x-[x]}}, & x>2\\ \mu, & x=2\end{cases}$$

where [x] is the greatest integer is than or equal to x. If f is continuous at x = 2, then $\lambda + \mu$ is equal to :

(A) e(−e + 1) (B) e(e − 2) (C) 1 (D) 2e − 1

Q28. MCQ (Single Correct Answer)

Open solution

The value of the definite integral $\int_{\pi/24}^{5\pi/24} \frac{dx}{1+\sqrt[3]{\tan 2x}}$ is :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{12}$ (D) $\frac{\pi}{18}$

Q29. MCQ (Single Correct Answer)

Open solution

If b is very small as compared to the value of a, so that the cube and other higher powers of $\frac{b}{a}$ can be neglected in the identity $\frac{1}{a-b}+\frac{1}{a-2b}+\frac{1}{a-3b}+\ldots..+\frac{1}{a-nb}=\alpha n+\beta n^2+\gamma n^3$, then the value of γ is :

(A) $\frac{a^2+b}{3a^3}$ (B) $\frac{a+b}{3a^2}$ (C) $\frac{b^2}{3a^3}$ (D) $\frac{a+b^2}{3a^3}$

Q30. MCQ (Single Correct Answer)

Open solution

Let y = y(x) be the solution of the differential equation $\frac{dy}{dx}=1+xe^{y-x}, -\sqrt{2}<x<\sqrt{2}, y(0)=0$

then, the minimum value of $y(x), x\in\left(-\sqrt{2},\sqrt{2}\right)$ is equal to :

(A) $\left(2-\sqrt{3}\right)-\log_e 2$ (B) $\left(2+\sqrt{3}\right)+\log_e 2$

(C) $\left(1+\sqrt{3}\right)-\log_e\left(\sqrt{3}-1\right)$ (D) $\left(1-\sqrt{3}\right)-\log_e\left(\sqrt{3}-1\right)$

Q31. MCQ (Single Correct Answer)

Open solution

The area (in sq. units) of the region, given by the set $\{(x,y)\in R\times R|x\geq 0, 2x^2\leq y\leq 4-2x\}$ is :

(A) $\frac{8}{3}$ (B) $\frac{17}{3}$ (C) $\frac{13}{3}$ (D) $\frac{7}{3}$

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Q32. MCQ (Single Correct Answer)

Open solution

Let g : N $\rightarrow$ N be defined as

g(3n + 1) = 3n + 2,

g(3n + 2) = 3n + 3,

g(3n + 3) = 3n + 1, for all n $\geq$ 0.

Then which of the following statements is true?

(A) There exists an onto function f : N $\rightarrow$ N such that fog = f

(B) There exists a one-one function f : N $\rightarrow$ N such that fog = f

(C) gogog = g

(D) There exists a function f : N $\rightarrow$ N such that gof = f

Q33. MCQ (Single Correct Answer)

Open solution

Let $f : [0, \infty) \rightarrow [0, \infty)$ be defined as $f(x) = \int_0^x [y]dy$

where [x] is the greatest integer less than or equal to x. Which of the following is true?

(A) f is continuous at every point in $[0, \infty)$ and differentiable except at the integer points.

(B) f is both continuous and differentiable except at the integer points in $[0, \infty)$.

(C) f is continuous everywhere except at the integer points in $[0, \infty)$.

(D) f is differentiable at every point in $[0, \infty)$.

Q34. MCQ (Single Correct Answer)

Open solution

The values of a and b, for which the system of equations

2x + 3y + 6z = 8

x + 2y + az = 5

3x + 5y + 9z = b

has no solution, are :

(A) a = 3, b $\neq$ 13 (B) a $\neq$ 3, b $\neq$ 13 (C) a $\neq$ 3, b = 3 (D) a = 3, b = 13

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Q35. MCQ (Single Correct Answer)

Let 9 distinct balls be distributed among 4 boxes, B_1, B_2, B_3 and B_4. If the probability that B_3 contains exactly 3 balls is $k\left(\frac{3}{4}\right)^9$ then k lies in the set :

(A) $\{x \in R : |x - 3| < 1\}$ (B) $\{x \in R : |x - 2| \leq 1\}$

(C) $\{x \in R : |x - 1| < 1\}$ (D) $\{x \in R : |x - 5| \leq 1\}$

Open solution

Q36. MCQ (Single Correct Answer)

The number of real roots of the equation $e^{6x} - e^{4x} - 2e^{3x} - 12e^{2x} + e^x + 1 = 0$ is :

(A) 2 (B) 4 (C) 6 (D) 1

Open solution

Q37. MCQ (Single Correct Answer)

Let an ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a^2 > b^2$, passes through $\left(\sqrt{\frac{3}{2}}, 1\right)$ and has eccentricity $\frac{1}{\sqrt{3}}$. If a circle, centered at focus F(α, 0), $\alpha > 0$, of E and radius $\frac{2}{\sqrt{3}}$, intersects E at two points P and Q, then PQ^2 is equal to :

(A) $\frac{8}{3}$ (B) $\frac{4}{3}$ (C) $\frac{16}{3}$ (D) 3

Open solution

Q38. Numerical

Let y = y(x) be solution of the following differential equation $e^y \frac{dy}{dx} - 2e^y \sin x + \sin x \cos^2 x = 0, y\left(\frac{\pi}{2}\right) = 0$ If $y(0) = \log_e(\alpha + \beta e^{-2})$, then $4(\alpha + \beta)$ is equal to _____.

Open solution

Q39. Numerical

Consider the following frequency distribution :

Class :	10-20	20-30	30-40	40-50	50-60
Frequency :	α	110	54	30	β

If the sum of all frequencies is 584 and median is 45, then $|\alpha - \beta|$ is equal to _____.

Open solution

Q40. Numerical

Let $\vec{p} = 2\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{q} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors. If a vector $\vec{r} = (\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k})$ is perpendicular to each of the vectors $(\vec{p} + \vec{q})$ and $(\vec{p} - \vec{q})$, and $|\vec{r}| = \sqrt{3}$, then $|\alpha| + |\beta| + |\gamma|$ is equal to _____.

Open solution

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Q41. Numerical

The ratio of the coefficient of the middle term in the expansion of $(1 + x)^{20}$ and the sum of the coefficients of two middle terms in expansion of $(1 + x)^{19}$ is _____.

Open solution

Q42. Numerical

There are 5 students in class 10, 6 students in class 11 and 8 students in class 12. If the number of ways, in which 10 students can be selected from them so as to include at least 2 students from each class and at most 5 students from the total 11 students of class 10 and 11 is 100 k, then k is equal to _____.

Open solution

Q43. Numerical

If α, β are roots of the equation $x^2 + 5(\sqrt{2})x + 10 = 0$, $\alpha > \beta$ and $P_n = \alpha^n - \beta^n$ for each positive integer n, then the value of $\left(\frac{P_{17}P_{20}+5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19}+5\sqrt{2}P_{18}^2}\right)$ is equal to _____.

Open solution

Q44. Numerical

The term independent of 'x' in the expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}}\right)^{10}$, where $x \neq 0, 1$ is equal to _____.

Open solution

Q45. Numerical

Let $S = \left\{n \in N \middle| \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}^n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \forall a, b, c, d \in R\right\}$, where $i = \sqrt{-1}$. Then the number of 2-digit numbers in the set S is _____.

Open solution

Physics 26 questions

Q46. MCQ (Single Correct Answer)

For a gas $C_P - C_V = R$ in a state P and $C_P - C_V = 1.10\ R$ in a state Q, T_P and T_Q are the temperatures in two different states P and Q respectively. Then

(A) $T_P = T_Q$ (B) $T_P < T_Q$ (C) $T_P = 0.9\ T_Q$ (D) $T_P > T_Q$

Open solution

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Q47. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Moment of inertia of a circular disc of mass 'M' and radius 'R' about X, Y axes (passing through its plane) and Z-axis which is perpendicular to its plane were found to be I_x, I_y and I_z respectively. The respectively radii of gyration about all the three axes will be the same.

Reason R : A rigid body making rotational motion has fixed mass and shape. In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both A and R are correct but R is NOT the correct explanation of A.

(B) A is not correct but R is correct.

(C) A is correct but R is not correct.

(D) Both A and R are correct and R is the correct explanation of A.

Q48. MCQ (Single Correct Answer)

Open solution

What should be the order of arrangement of de-Broglie wavelength of electron (λ_e), an α-particle (λ_a) and proton (λ_p) given that all have the same kinetic energy?

(A) $\lambda_e = \lambda_p = \lambda_\alpha$

(B) $\lambda_e < \lambda_p < \lambda_\alpha$

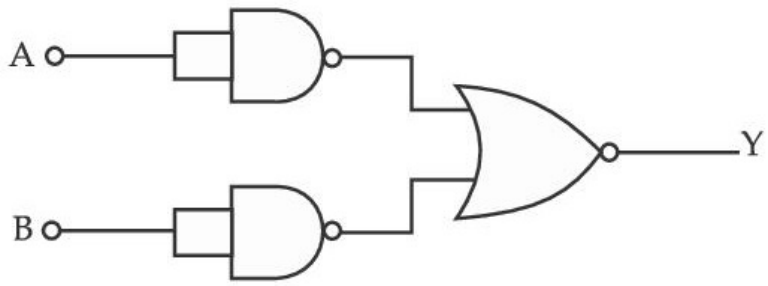
(C) $\lambda_e > \lambda_p > \lambda_\alpha$

(D) $\lambda_e = \lambda_p > \lambda_\alpha$

Q49. MCQ (Single Correct Answer)

Open solution

Identify the logic operation carried out.



(A) OR

(B) AND

(C) NOR

(D) NAND

Q50. MCQ (Single Correct Answer)

Open solution

A particle of mass 4M at rest disintegrates into two particles of mass M and 3M respectively having non zero velocities. The ratio of de-Broglie wavelength of particle of mass M to that of mass 3M will be :

(A) 1 : 3

(B) 3 : 1

(C) $1 : \sqrt{3}$

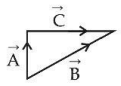
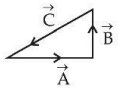
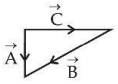
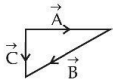
(D) 1 : 1

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Q51. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II

List I	List II
(a) $\vec{C}-\vec{A}-\vec{B}=0$	(i) 
(b) $\vec{A}-\vec{C}-\vec{B}=0$	(ii) 
(c) $\vec{B}-\vec{A}-\vec{C}=0$	(iii) 
(d) $\vec{A}+\vec{B}=-\vec{C}$	(iv) 

Choose the correct answer from the options given below :

(A) (a) → (iv), (b) → (i), (c) → (iii), (d) → (ii)

(B) (a) → (iv), (b) → (iii), (c) → (i), (d) → (ii)

(C) (a) → (iii), (b) → (ii), (c) → (iv), (d) → (i)

(D) (a) → (i), (b) → (iv), (c) → (ii), (d) → (iii)

Q52. MCQ (Single Correct Answer)

Open solution

A parallel plate capacitor with plate area 'A' and distance of separation 'd' is filled with a dielectric. What is the capacity of the capacitor when permittivity of the dielectric varies as :

$\varepsilon(x)=\varepsilon_0+kx$, for $\left(0<x\leq\frac{d}{2}\right)$

$\varepsilon(x)=\varepsilon_0+k(d-x)$, for $\left(\frac{d}{2}\leq x\leq d\right)$

(A) $\left(\varepsilon_0+\frac{kd}{2}\right)^{2/kA}$

(B) $\frac{kA}{2\ln\left(\frac{2\varepsilon_0+kd}{2\varepsilon_0}\right)}$

(C) 0

(D) $\frac{kA}{2}\ln\left(\frac{2\varepsilon_0}{2\varepsilon_0-kd}\right)$

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Q53. MCQ (Single Correct Answer)

Open solution

A monoatomic ideal gas, initially at temperature T_1 is enclosed in a cylinder fitted with a frictionless piston. The gas is allowed to expand adiabatically to a temperature T_2 by releasing the piston suddenly. If l_1 and l_2 are the lengths of the gas column, before and after the expansion respectively, then the value of $\frac{T_1}{T_2}$ will be :

(A) $\left(\frac{l_1}{l_2}\right)^{\frac{2}{3}}$

(B) $\left(\frac{l_2}{l_1}\right)^{\frac{2}{3}}$

(C) $\frac{l_2}{l_1}$

(D) $\frac{l_1}{l_2}$

Q54. MCQ (Single Correct Answer)

Open solution

A ray of laser of a wavelength 630 nm is incident at an angle of 30° at the diamond-air interface. It is going from diamond to air. The refractive index of diamond is 2.42 and that of air is 1. Choose the correct option.

(A) angle of refraction is 24.41°

(B) angle of refraction is 30°

(C) refraction is not possible

(D) angle of refraction is 53.4°

Q55. MCQ (Single Correct Answer)

Open solution

Two wires of same length and radius are joined end to end and loaded. The Young's moduli of the materials of the two wires are Y_1 and Y_2. The combination behaves as a single wire then its Young's modulus is :

(A) $Y = \frac{2Y_1Y_2}{3(Y_1+Y_2)}$

(B) $Y = \frac{2Y_1Y_2}{Y_1+Y_2}$

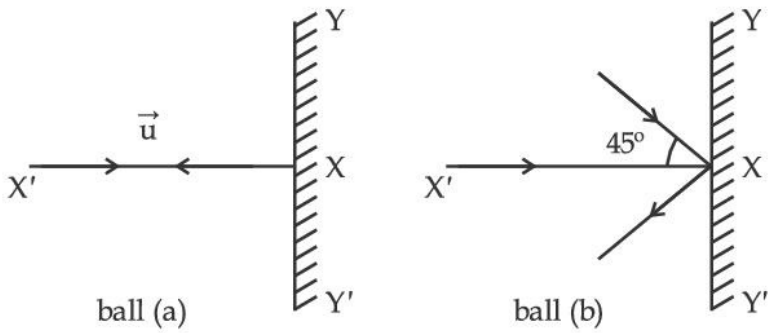
(C) $Y = \frac{Y_1Y_2}{2(Y_1+Y_2)}$

(D) $Y = \frac{Y_1Y_2}{Y_1+Y_2}$

Q56. MCQ (Single Correct Answer)

Open solution

Two billiard balls of equal mass 30g strike a rigid wall with same speed of 108 kmph (as shown) but at different angles. If the balls get reflected with the same speed then the ratio of the magnitude of impulses imparted to ball 'a' and ball 'b' by the wall along 'X' direction is :



(A) 1 : 1

(B) $\sqrt{2}$: 1

(C) 2 : 1

(D) 1 : $\sqrt{2}$

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Q57. MCQ (Single Correct Answer)

Open solution

In the Young's double slit experiment, the distance between the slits varies in time as $d(t) = d_0 + a_0 \sin\omega t$; where d_0, ω and a_0 are constants. The difference between the largest fringe width and the smallest fringe width obtained over time is given as :

(A) $\frac{2\lambda D(d_0)}{(d_0^2 - a_0^2)}$

(B) $\frac{2\lambda D a_0}{(d_0^2 - a_0^2)}$

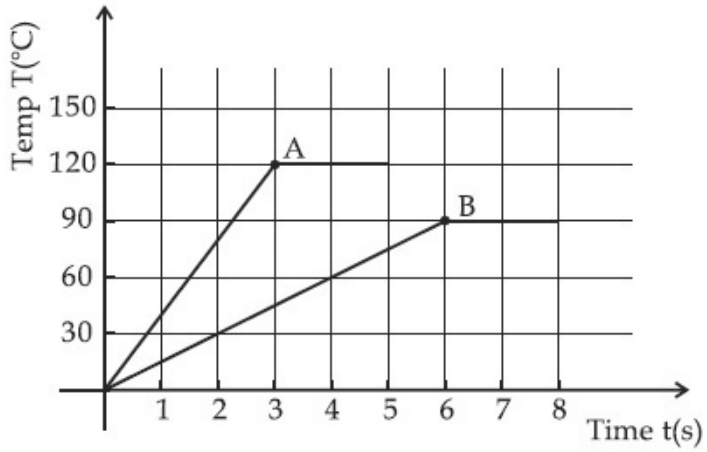
(C) $\frac{\lambda D}{d_0^2} a_0$

(D) $\frac{\lambda D}{d_0 + a_0}$

Q58. MCQ (Single Correct Answer)

Open solution

Two different metal bodies A and B of equal mass are heated at a uniform rate under similar conditions. The variation of temperature of the bodies is graphically represented as shown in the figure. The ratio of specific heat capacities is :



(A) $\frac{8}{3}$

(B) $\frac{3}{8}$

(C) $\frac{3}{4}$

(D) $\frac{4}{3}$

Q59. MCQ (Single Correct Answer)

Open solution

A linearly polarized electromagnetic wave in vacuum

$$E = 3.1 \cos\left[(1.8)z - (5.4 \times 10^6)t\right]\hat{i}N/C$$

is incident normally on a perfectly reflecting wall at z = a. Choose the correct option

(A) The wavelength is 5.4 m

(B) The frequency of electromagnetic wave is 54×10^4 Hz.

(C) The transmitted wave will be $3.1 \cos\left[(1.8)z - (5.4 \times 10^6)t\right]\hat{i}N/C$

(D) The reflected wave will be $3.1 \cos\left[(1.8)z + (5.4 \times 10^6)t\right]\hat{i}N/C$

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Q60. MCQ (Single Correct Answer)

Open solution

Water droplets are coming from an open tap at a particular rate. The spacing between a droplet observed at 4th second after its fall to the next droplet is 34.3 m. At what rate the droplets are coming from the tap ? (Take g = 9.8 m/s^2)

(A) 3 drops / 2 seconds

(B) 2 drops / second

(C) 1 drop / second

(D) 1 drop / 7 seconds

Q61. MCQ (Single Correct Answer)

Open solution

The minimum and maximum distances of a planet revolving around the sun are x_1 and x_2. If the minimum speed of the planet on its trajectory is v_0 then its maximum speed will be :

(A) $\frac{v_0 x_1^2}{x_2^2}$ (B) $\frac{v_0 x_2^2}{x_1^2}$ (C) $\frac{v_0 x_1}{x_2}$ (D) $\frac{v_0 x_2}{x_1}$

Q62. Numerical

Open solution

A body of mass 2 kg moving with a speed of 4 m/s makes an elastic collision with another body at rest and continues to move in the original direction but with one fourth of its initial speed. The speed of the two body centre of mass is $\frac{x}{10}$ m/s. Then the value of x is _____.

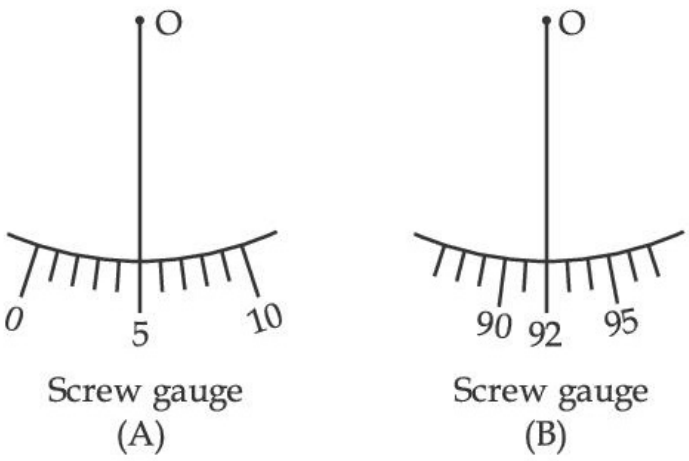
Q63. Numerical

Open solution

Student A and student B used two screw gauges of equal pitch and 100 equal circular divisions to measure the radius of a given wire. The actual value of the radius of the wire is 0.322 cm. The absolute value of the difference between the final circular scale readings observed by the students A and B is _____.

[Figure shows position of reference 'O' when jaws of screw gauge are closed]

Given pitch = 0.1 cm.



Screw gauge (A)

Screw gauge (B)

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Q64. Numerical

An inductor of 10 mH is connected to a 20V battery through a resistor of 10 kΩ and a switch. After a long time, when maximum current is set up in the circuit, the current is switched off. The current in the circuit after 1 μs is $\frac{x}{100}$ mA. Then x is equal to _____. (Take e^{-1} = 0.37)

Open solution

Q65. Numerical

A circular conducting coil of radius 1 m is being heated by the change of magnetic field $\vec{B}$ passing perpendicular to the plane in which the coil is laid. The resistance of the coil is 2 $\mu\Omega$. The magnetic field is slowly switched off such that its magnitude changes in time as

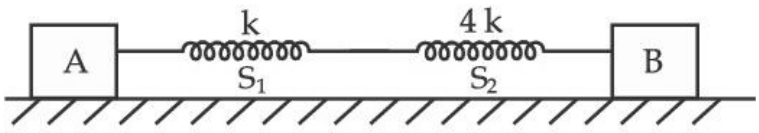
$B = \frac{4}{\pi} \times 10^{-3} T \left(1 - \frac{t}{100}\right)$

The energy dissipated by the coil before the magnetic field is switched off completely is E = _____ mJ.

Open solution

Q66. Numerical

In the reported figure, two bodies A and B of masses 200 g and 800 g are attached with the system of springs. Springs are kept in a stretched position with some extension when the system is released. The horizontal surface is assumed to be frictionless. The angular frequency will be _____ rad/s when k = 20 N/m.



Open solution

Q67. Numerical

The value of aluminium susceptibility is 2.2×10^{-5}. The percentage increase in the magnetic field if space within a current carrying toroid is filled with aluminium is $\frac{x}{10^4}$. Then the value of x is _____.

Open solution

Q68. Numerical

A particle of mass 1 mg and charge q is lying at the mid-point of two stationary particles kept at a distance '2 m' when each is carrying same charge 'q'. If the free charged particle is displaced from its equilibrium position through distance 'x' (x << 1 m), the particle executes SHM. Its angular frequency of oscillation will be _____ $\times 10^5$ rad/s if q^2 = 10 C^2.

Open solution

Q69. Numerical

An electric bulb rated as 200 W at 100 V is used in a circuit having 200 V supply. The resistance 'R' that must be put in series with the bulb so that the bulb delivers the same power is _____ Ω.

Open solution

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Q70. Numerical

A pendulum bob has a speed of 3 m/s at its lowest position. The pendulum is 50 cm long. The speed of bob, when the length makes an angle of 60° to the vertical will be (g = 10 m/s^2) _____ m/s.

Open solution

Q71. Numerical

A particle of mass 'm' is moving in time 't' on a trajectory given by

$$\vec{r} = 10\alpha t^2\hat{i} + 5\beta(t-5)\hat{j}$$

Where α and β are dimensional constants.

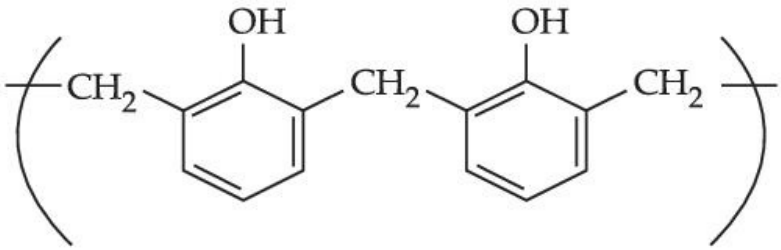
The angular momentum of the particle becomes the same as it was for t = 0 at time t = _____ seconds.

Open solution

Out of syllabus 19 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q72. MCQ (Single Correct Answer) · **Chemistry**



is a repeating unit for :

(A) Novolac (B) Buna-N (C) Acrilan (D) Neoprene

Open solution

Q73. MCQ (Single Correct Answer) · **Chemistry**

Sodium stearate $CH_3(CH_2)_{16}COO^-Na^+$ is an anionic surfactant which forms micelles in oil.

Choose the correct statement for it from the following :

(A) It forms spherical micelles with $CH_3(CH_2)_{16}$ - group pointing towards the centre of sphere.

(B) It forms non-spherical micelles with -COO^- group pointing outwards on the surface.

(C) It forms spherical micelles with $CH_3(CH_2)_{16}$ - group pointing outwards on the surface of sphere

(D) It forms non-spherical micelles with $CH_3(CH_2)_{16}$ - group pointing towards the centre.

Open solution

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Q74. MCQ (Single Correct Answer) · **Chemistry**

At 298.2 K the relationship between enthalpy of bond dissociation (in kJ mol^{-1}) for hydrogen (E_H) and its isotope, deuterium (E_D), is best described by :

(A) $E_H = \frac{1}{2}E_D$ (B) $E_H = E_D$ (C) $E_H \simeq E_D - 7.5$ (D) $E_H = 2E_D$

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements :

Statement I : None of the alkaline earth metal hydroxides dissolve in alkali.

Statement II : Solubility of alkaline earth metal hydroxides in water increases down the group.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but Statement II is incorrect.

(B) Statement I is incorrect but Statement II is correct.

(C) Statement I and Statement II both are incorrect.

(D) Statement I and Statement II both are correct.

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following chemical agent is not being used for dry-cleaning of clothes?

(A) H_2O_2 (B) CCl_4 (C) Liquid CO_2 (D) $Cl_2C = CCl_2$

Open solution

Q77. MCQ (Single Correct Answer) · **Chemistry**

In the leaching of alumina from bauxite, the ore expected to leach out in the process by reacting with NaOH is :

(A) TiO_2 (B) Fe_2O_3 (C) ZnO (D) SiO_2

Open solution

Q78. Numerical · **Chemistry**

A home owner uses 4.00×10^3 m^3 of methane (CH_4) gas, (assume CH_4 is an ideal gas) in a year to heat his home. Under the pressure of 1.0 atm and 300 K, mass of gas used is $x \times 10^5$ g. The value of x is _____. (Nearest integer)

(Given R = 0.083 L atm K^{-1} mol^{-1})

Open solution

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Q79. MCQ (Single Correct Answer) · **Mathematics**

A spherical gas balloon of radius 16 meter subtends an angle 60° at the eye of the observer A while the angle of elevation of its center from the eye of A is 75°. Then the height (in meter) of the top most point of the balloon from the level of the observer's eye is :

Open solution

(A) $8(2+2\sqrt{3}+\sqrt{2})$ (B) $8(\sqrt{6}+\sqrt{2}+2)$ (C) $8(\sqrt{2}+2+\sqrt{3})$ (D) $8(\sqrt{6}-\sqrt{2}+2)$

Q80. MCQ (Single Correct Answer) · **Mathematics**

Let the vectors

$(2+a+b)\hat{i}+(a+2b+c)\hat{j}-(b+c)\hat{k}, (1+b)\hat{i}+2b\hat{j}-b\hat{k}$ and $(2+b)\hat{i}+2b\hat{j}+(1-b)\hat{k}, a,b,c\in R$

be co-planar. Then which of the following is true?

Open solution

(A) 2b = a + c (B) 3c = a + b (C) a = b + 2c (D) 2a = b + c

Q81. MCQ (Single Correct Answer) · **Mathematics**

The Boolean expression $(p\Rightarrow q)\wedge(q\Rightarrow\sim p)$ is equivalent to :

Open solution

(A) $\sim$ q (B) q (C) p (D) $\sim$ p

Q82. MCQ (Single Correct Answer) · **Mathematics**

The sum of all values of x in [0, 2π], for which sin x + sin 2x + sin 3x + sin 4x = 0, is equal to :

Open solution

(A) 8π (B) 11π (C) 12π (D) 9π

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let a parabola P be such that its vertex and focus lie on the positive x-axis at a distance 2 and 4 units from the origin, respectively. If tangents are drawn from O(0, 0) to the parabola P which meet P at S and R, then the area (in sq. units) of ΔSOR is equal to :

Open solution

(A) $16\sqrt{2}$ (B) 16 (C) 32 (D) $8\sqrt{2}$

Q84. MCQ (Single Correct Answer) · **Mathematics**

Let the foot of perpendicular from a point P(1, 2, -1) to the straight line $L:\frac{x}{1}=\frac{y}{0}=\frac{z}{-1}$ be N. Let a line be drawn from P parallel to the plane x + y + 2z = 0 which meets L at point Q. If α is the acute angle between the lines PN and PQ, then cosα is equal to _____.

Open solution

(A) $\frac{1}{\sqrt{5}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{2\sqrt{3}}$

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Q85. Numerical · **Mathematics**

If the value of

$$\left(1+\frac{2}{3}+\frac{6}{3^2}+\frac{10}{3^3}+\ldots . upto\ \infty\right)^{\log_{(0.25)}\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\ldots . upto\ \infty\right)}$$

is l, then l^2 is equal to _____.

Open solution

Q86. Numerical · **Mathematics**

Let $M=\left\{A=\begin{pmatrix}a & b\\ c & d\end{pmatrix} : a,b,c,d \in \{\pm 3,\pm 2,\pm 1,0\}\right\}$. Define f : M $\rightarrow$ Z, as f(A) = det(A), for all A$\in$M, where z is set of all integers. Then the number of A$\in$M such that f(A) = 15 is equal to _____.

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

Some nuclei of a radioactive material are undergoing radioactive decay. The time gap between the instances when a quarter of the nuclei have decayed and when half of the nuclei have decayed is given as :

(where λ is the decay constant)

(A) $\frac{1}{2}\frac{\ln 2}{\lambda}$

(B) $\frac{\ln 2}{\lambda}$

(C) $\frac{2\ln 2}{\lambda}$

(D) $\frac{\ln \frac{3}{2}}{\lambda}$

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

The half-life of ^{198}Au is 3 days. If atomic weight of ^{198}Au is 198 g/mol then the activity of 2 mg of ^{198}Au is [in disintegration/second] :

(A) 2.67×10^{12}

(B) 6.06×10^{18}

(C) 32.36×10^{12}

(D) 16.18×10^{12}

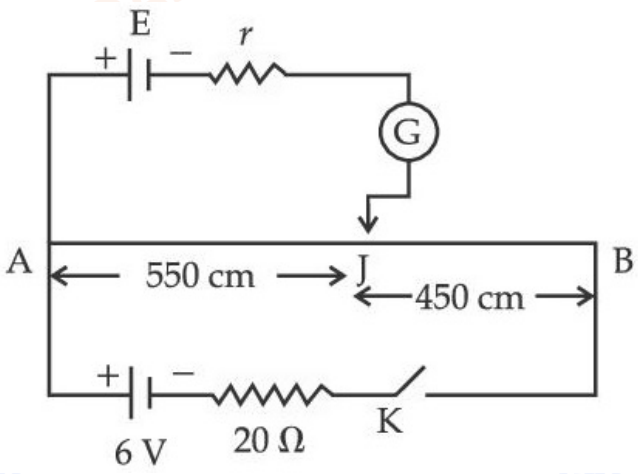
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Q89. MCQ (Single Correct Answer) · **Physics**

Open solution

In the given figure, there is a circuit of potentiometer of length AB = 10 m. The resistance per unit length is 0.1 Ω per cm. Across AB, a battery of emf E and internal resistance 'r' is connected. The maximum value of emf measured by this potentiometer is :



(A) 5 V (B) 2.25 V (C) 6 V (D) 2.75 V

Q90. MCQ (Single Correct Answer) · **Physics**

Open solution

In amplitude modulation, the message signal

$V_m(t) = 10 \sin (2\pi \times 10^5 t)$ volts and

Carrier signal

$V_C(t) = 20 \sin(2\pi \times 10^7 t)$ volts

The modulated signal now contains the message signal with lower side band and upper side band frequency, therefore the bandwidth of modulated signal is α kHz. The value of α is :

(A) 200 kHz (B) 50 kHz (C) 100 kHz (D) 0

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Answer Key

Chemistry

1. **A**	2. **C**	3. **B**	4. **D**	5. **C**
6. **C**	7. **B**	8. **B**	9. **C**	10. **C**
11. **A**	12. **C**	13. **A**	14. **A**	15. **10**
16. **0**	17. **2**	18. **25**	19. **25**	20. **24**
21. **101**	22. **18**	23. **464**		

Mathematics

24. **D**	25. **A**	26. **A**	27. **A**	28. **C**
29. **C**	30. **D**	31. **D**	32. **A**	33. **A**
34. **A**	35. **A**	36. **A**	37. **C**	38. **4**
39. **164**	40. **3**	41. **1**	42. **238**	43. **1**
44. **210**	45. **11**			

Physics

46. **D**	47. **B**	48. **C**	49. **B**	50. **D**
51. **B**	52. **B**	53. **B**	54. **C**	55. **B**
56. **B**	57. **B**	58. **B**	59. **D**	60. **C**
61. **D**	62. **25**	63. **13**	64. **74**	65. **80**
66. **10**	67. **22**	68. **6000**	69. **50**	70. **2**
71. **10**				

Out of syllabus

72. **A**	73. **A**	74. **C**	75. **B**	76. **A**
77. **D**	78. **26**	79. **B**	80. **A**	81. **D**
82. **D**	83. **B**	84. **C**	85. **3**	86. **16**
87. **D**	88. **D**	89. **A**	90. **A**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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